

Complex Dynamics and High codimension Bifurcations of a Predator Prey Model

Wael El Khateeb

University of Toledo

In collaboration with Dr. Chunhua Shan, Dr. Guihong Fan, and Dr. Hao Wang.



History



Adriatic Sea: East of Italy

During World War I, fishing in regions such as the Adriatic Sea declined dramatically, leading to unusual fluctuations in fish populations. Vito Volterra became intrigued by this phenomenon.

Fluctuation in population numbers

Fishermen noticed that the populations of certain prey fish and their predators did not simply increase or decrease together; instead, they appeared to oscillate in a regular, cyclical fashion.



Florida Insects

<http://bit.ly/4hpAqiz>

Real-Life Data



Classical Model

The Lotka–Volterra model is a pair of first-order, nonlinear differential equations that describe the dynamics of predator and prey populations. Let $x(t)$ represent the prey population and $y(t)$ represent the predator population. The model is given by:

$$\begin{cases} \frac{dx}{dt} = \alpha x - \beta xy, \\ \frac{dy}{dt} = \delta xy - \gamma y, \end{cases} \quad (1)$$

where:

- α is the natural growth rate of the prey,
- β is the predation rate coefficient,
- δ is the growth rate of predators per prey consumed, and
- γ is the natural death rate of the predators.

Extension of Classical Model

Instead of exponential growth, logistic growth is used to account for resource limitations:

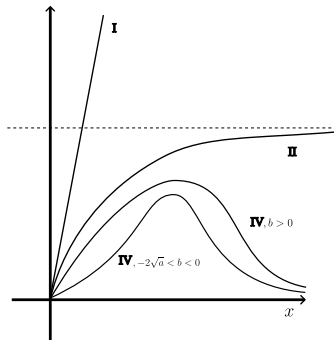
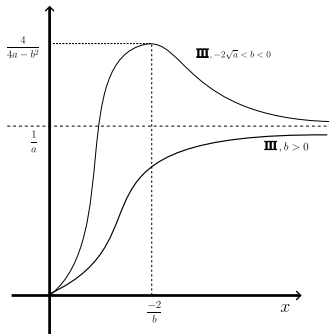
$$\frac{dx}{dt} = \alpha x \left(1 - \frac{x}{K}\right) - p(x)y,$$

where K is the carrying capacity. We can incorporate the natural harvesting of prey by subtracting h :

$$\frac{dx}{dt} = \alpha x(t) \left(1 - \frac{x(t)}{K}\right) - p(x)y - h.$$

Holling Functional Response

Describe how a predator's rate of prey consumption changes with prey density. They were developed by Canadian ecologist C. S. Holling in the 1950s and 1960s.



Linearization-ODE 1

Definition (Fixed Point)

A point (x^*, y^*) is called a *fixed point* of the dynamical system

$$\begin{cases} \dot{x} = f(x, y) \\ \dot{y} = g(x, y) \end{cases}$$

if it satisfies $f(x^*, y^*) = g(x^*, y^*) = 0$

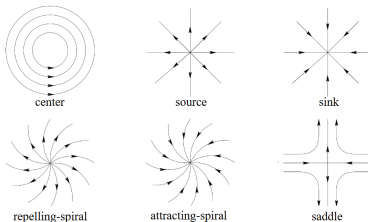
Theorem (Hartman-Grobman)

Near a hyperbolic fixed point E , the dynamics of a nonlinear system are topologically conjugate to its linearization.

Linearization ODE 2

$$\begin{cases} \frac{dx}{dt} = f(x, y), \\ \frac{dy}{dt} = g(x, y), \end{cases} \rightarrow J = \begin{bmatrix} f_x(x^*, y^*) & f_y(x^*, y^*) \\ g_x(x^*, y^*) & g_y(x^*, y^*) \end{bmatrix} \rightarrow \lambda^2 - \text{Tr}(J)\lambda + \det(J) = 0$$

Based on the eigenvalues, we can deduce the type of equilibria.



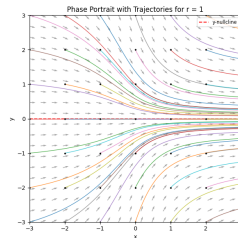
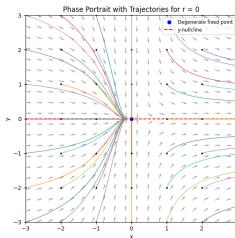
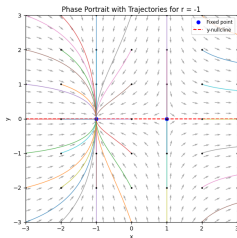
VERMA V., KAO D., PANG A.: A flow-guided streamline seeding strategy. In Proceedings of the conference on Visualization'00 (2000), IEEE Computer Society Press, pp. 163–170

Codimension 1 Bifurcations

Definition (Bifurcation)

qualitative change in the behavior of a system as a parameter is varied.

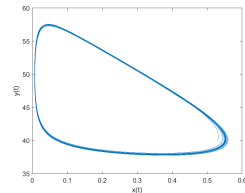
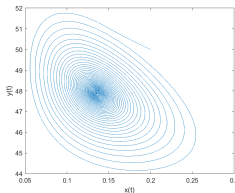
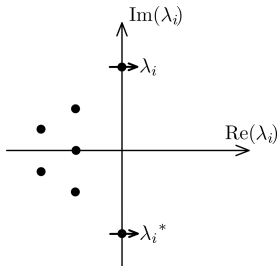
$$\begin{cases} \frac{dx}{dt} = r + x^2, \\ \frac{dy}{dt} = -y. \end{cases} \quad (2)$$



Local Hopf Bifurcation

Definition

A *Hopf bifurcation* occurs when a fixed point of a dynamical system loses stability as a pair of complex conjugate eigenvalues crosses the imaginary axis, resulting in the emergence or disappearance of a periodic orbit of small amplitude.



Translation of the predation action

Predators need time to reproduce after predation, which is why we add a delay.

$$\begin{cases} \frac{dx}{dt} = rx(1 - \frac{x}{K}) - p(x(t - \tau))y, \\ \frac{dy}{dt} = p(x(t))y - dy. \end{cases} \quad (3)$$

τ represents the time required for the predator's consumption of prey to translate into an increase in the predator population. The mortality of predators during this period is accounted for by the function $g(\tau, d)$:

$$\begin{cases} \frac{dx}{dt} = rx(1 - \frac{x}{K}) - p(x(t))y, \\ \frac{dy}{dt} = g(\tau, d)p(x(t - \tau))y - dy. \end{cases} \quad (4)$$

Functional Differential Equation

Define

$$C = C([-r, 0], \mathbb{R}^n)$$

Let

$$\sigma \in \mathbb{R}, r, A \in \mathbb{R}^+, x \in C([\sigma - r, \sigma + A], \mathbb{R}^n), \text{ and } t \in [\sigma, \sigma + A]$$

We define $x_t(\theta) = x(t + \theta)$, $\theta \in [-r, 0]$. Assume Ω is a subset of $\mathbb{R} \times C$, $f: \Omega \rightarrow \mathbb{R}^n$ is a given function, then we call:

$$\dot{x}(t) = f(t, x_t) \tag{5}$$

a retarded delay differential equation or functional differential equation. Initial data for the delay differential equations are generally continuous functions defined on a finite interval.

Existence and Uniqueness of Solutions

Theorem (Existence)

Suppose that Ω is an open subset in $\mathbb{R} \times C$ and f is continuous on Ω . If $(\sigma, \phi) \in \Omega$, then there is a solution of (1.1) passing through (σ, ϕ) .

Theorem (Uniqueness)

Suppose Ω is an open set in $\mathbb{R} \times C$, $f: \Omega \rightarrow \mathbb{R}^n$ is continuous and $f(t, \phi)$ is Lipschitz in ϕ in each compact set in Ω . If $(\sigma, \phi) \in \Omega$, then there is a unique solution of Equation 5 through (σ, ϕ)

Note: We say $f(t, \phi)$ is Lipschitz in ϕ in a compact set K of $\mathbb{R} \times C$ if there is a constant $k > 0$ such that for any $(t, \phi_i) \in K, i = 1, 2$,

$$|f(t, \phi_1) - f(t, \phi_2)| \leq k|\phi_1 - \phi_2|$$

Linearization DDE 1

$$\begin{cases} \frac{dx}{dt} = f(x, y, x(t-\tau), y(t-\tau)) \\ \frac{dy}{dt} = g(x, y, x(t-\tau), y(t-\tau)) \end{cases} \rightarrow \lambda^2 + C_1\lambda + e^{-\lambda\tau}(C_2\lambda + C_3) + C_4 = 0$$

Possibly infinitely many eigenvalues satisfy the characteristics equation.

Finding Bifurcation points

We are interested in finding the delay thresholds that get us pure imaginary eigenvalues, ie, $\lambda = i\omega$. We replace λ by $i\omega$ in the characteristics equation:

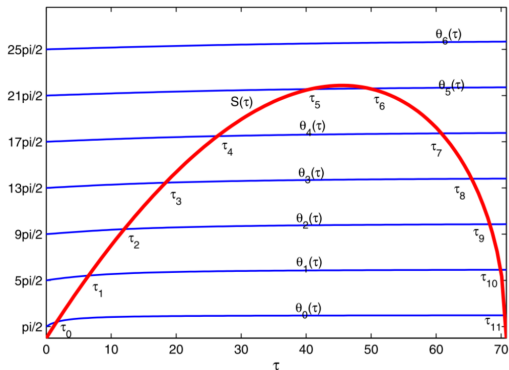
$$\lambda^2 + C_1\lambda + e^{-\lambda\tau}(C_2\lambda + C_3) + C_4 = 0 \rightarrow -\omega^2 + C_1i\omega + e^{-i\omega\tau}(C_2i\omega + C_3) + C_4 = 0.$$

Using Euler's Formula and the trigonometric identity, we get:

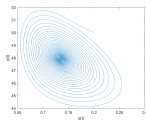
$$\begin{cases} \cos(\omega\tau) = -\frac{C_1C_2\omega^2 - C_3\omega^2 + C_3C_4}{C_2^2\omega^2 + C_3^2}, \\ \sin(\omega\tau) = \frac{\omega(C_2\omega^2 + C_1C_3 - C_2C_4)}{C_2^2\omega^2 + C_3^2}, \end{cases} \quad \text{and} \quad \frac{\omega^4 + (C_1^2 - 2C_4)\omega^2 + C_4^2}{C_2^2\omega^2 + C_3^2} = 1.$$

Finding Bifurcation points 2

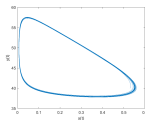
Accordingly, we are interested in finding the points (ω, τ) that satisfy both equations!



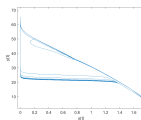
Global Hopf Bifurcation



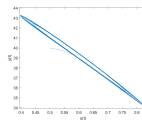
$\tau = 0.5$



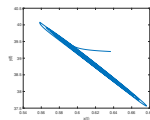
$\tau = 0.7$



$\tau = 8.5$

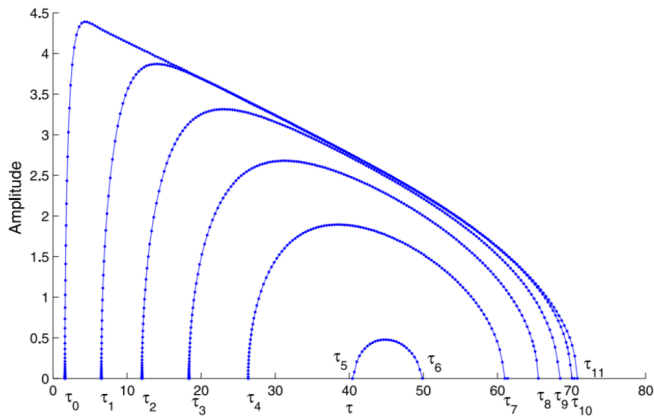


$\tau = 14.4$

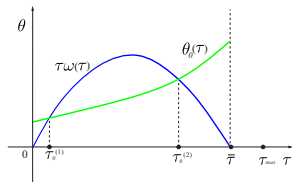


$\tau = 14.8$

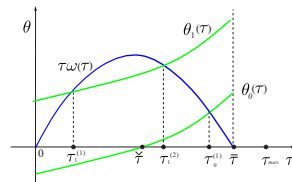
Global Hopf Bifurcation 2



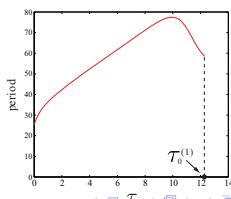
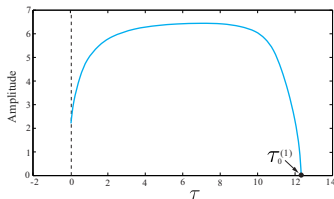
Novel Results 1



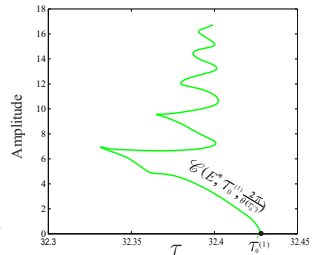
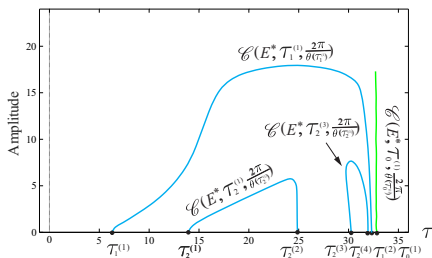
$K < K_0$



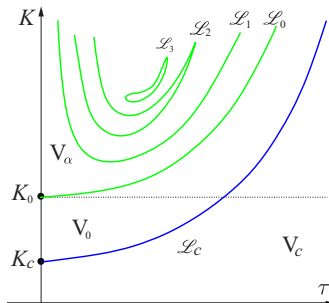
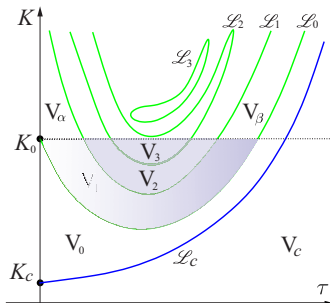
$K > K_0$



Novel Results 2



Novel Results 3



Codimension n Bifurcation

Definition (Codimension n Bifurcation)

Bifurcation that requires the variation of n independent parameters $(\beta_1, \beta_2, \dots, \beta_n)$ for its characteristic degeneracies to unfold.

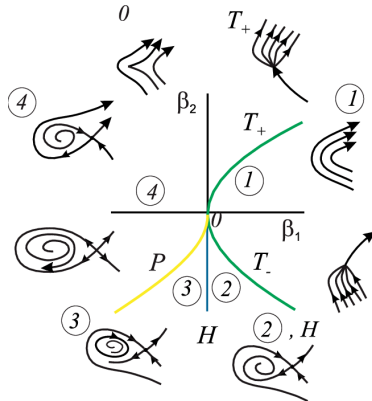
Normal Forms and Codimension 2 Bogdanov-Takens Bifurcation

Definition (Normal Form Theory)

A technique for transforming ordinary differential equations that describe nonlinear dynamical systems into standard forms using near-identity transformations. This process eliminates non-resonant nonlinear terms through a change of coordinates while preserving the essential dynamics.

$$\begin{cases} \frac{dx}{dt} = f(x, y, \beta_1, \beta_2) \\ \frac{dy}{dt} = g(x, y, \beta_1, \beta_2) \end{cases} \xrightarrow{C^\infty \text{ Diffeomorphism}} \begin{cases} \frac{dx}{dt} = y \\ \frac{dy}{dt} = \beta_1 + \beta_2 x + x^2 + xy + O(3) \end{cases}$$

Codimension 2 BT unfolding



ANALYSIS OF THE TAKENS–BOGDANOV BIFURCATION ON m -PARAMETERIZED VECTOR FIELDS -
Scientific Figure on ResearchGate.

Codimension 3 BT bifurcation

The normal form for a codimension-3 Bogdanov-Takens bifurcation is given by:

$$\begin{cases} \dot{x} = y, \\ \dot{y} = \beta_1 + \beta_2 x + \beta_3 y + x^2 + xy + \alpha x^3. \end{cases} \quad (6)$$

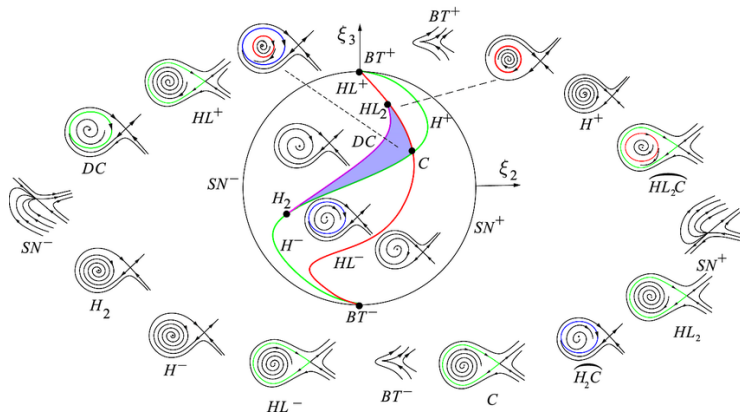
where:

- $\beta_1, \beta_2, \beta_3$ are unfolding parameters,
- α is a higher-order nonlinearity parameter.

This system exhibits:

- Saddle-node bifurcation of equilibria,
- Homoclinic bifurcation,
- Hopf bifurcation leading to limit cycles,
- Cusped homoclinic bifurcation.

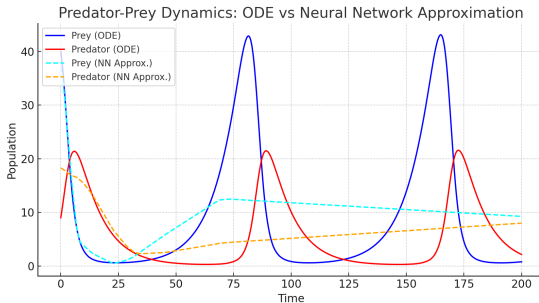
BT3 Diagram



Arsie, A., Kottegoda, C. Shan, C. High Codimension Bifurcations of a Predator–Prey System with Generalized Holling Type III Functional Response and Allee Effects. J Dyn Diff Equat 35, 3355–3380 (2023).
<https://doi.org/10.1007/s10884-022-10214-6>

Case Study: Predator-Prey Model (Lotka-Volterra)

- The NN fails to capture the periodic oscillations over long time.
- **ODEs provide physically meaningful and accurate long-term dynamics**, while NNs struggle with generalization beyond training data.



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